F3XP M.A.R.S.

A Level Physics Practice Paper 1

CANDIDATE NAME

CENTRE NUMBER | | | | | |

CANDIDATE NUMBER | | | | |

2999792458

PHYSICS **9702/41**

Paper 4 A Level Structured Questions **August/September 2026**

2 hours

You must answer on the question paper.

No additional materials are needed.

INSTRUCTIONS

- Answer **all** questions.
- Use a black or dark blue pen. You may use an HB pencil for any diagrams or graphs.
- Write your name, centre number and candidate number in the boxes at the top of the page.
- Write your answer to each question in the space provided.
- Do **not** use an erasable pen or correction fluid.
- You may use a calculator.
- You should show all your working and use appropriate units.

INFORMATION

- The total mark for this paper is 100.
- The number of marks for each question or part question is shown in brackets [ ].

This document has **24** pages. Any blank pages are indicated.

Data

acceleration of free fall	$g = 9.81\ \text{m s}^{-2}$
speed of light in free space	$c = 3.00 \times 10^{8}\ \text{m s}^{-1}$
elementary charge	$e = 1.60 \times 10^{-19}\ \text{C}$
unified atomic mass unit	$1\ \text{u} = 1.66 \times 10^{-27}\ \text{kg}$
rest mass of proton	$m_\text{p} = 1.67 \times 10^{-27}\ \text{kg}$
rest mass of electron	$m_\text{e} = 9.11 \times 10^{-31}\ \text{kg}$
Avogadro constant	$N_\text{A} = 6.02 \times 10^{23}\ \text{mol}^{-1}$
molar gas constant	$R = 8.31\ \text{J K}^{-1}\ \text{mol}^{-1}$
Boltzmann constant	$k = 1.38 \times 10^{-23}\ \text{J K}^{-1}$
gravitational constant	$G = 6.67 \times 10^{-11}\ \text{N m}^2\ \text{kg}^{-2}$
permittivity of free space	$\varepsilon_0 = 8.85 \times 10^{-12}\ \text{F m}^{-1}$ $\left(\frac{1}{4\pi\varepsilon_0} = 8.99 \times 10^{9}\ \text{m F}^{-1}\right)$
Planck constant	$h = 6.63 \times 10^{-34}\ \text{J s}$
Stefan–Boltzmann constant	$\sigma = 5.67 \times 10^{-8}\ \text{W m}^{-2}\ \text{K}^{-4}$

Formulae

uniformly accelerated motion	$s = ut + \frac{1}{2}at^2$ $v^2 = u^2 + 2as$
hydrostatic pressure	$\Delta p = \rho g \Delta h$
upthrust	$F = \rho g V$
Doppler effect for sound waves	$f_\text{o} = \frac{f_\text{s} v}{v \pm v_\text{s}}$
electric current	$I = Anvq$
resistors in series	$R = R_1 + R_2 + ...$
resistors in parallel	$\frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2} + ...$

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gravitational potential	$\phi = -\frac{GM}{r}$
gravitational potential energy	$E_P = -\frac{GMm}{r}$
pressure of an ideal gas	$p = \frac{1}{3}\frac{Nm}{V}\langle c^2 \rangle$
simple harmonic motion	$a = -\omega^2 x$
velocity of particle in s.h.m.	$v = v_0 \cos \omega t$ $v = \pm \omega \sqrt{(x_0^2 - x^2)}$
electric potential	$V = \frac{Q}{4\pi\varepsilon_0 r}$
electrical potential energy	$E_P = \frac{Qq}{4\pi\varepsilon_0 r}$
capacitors in series	$\frac{1}{C} = \frac{1}{C_1} + \frac{1}{C_2} + ...$
capacitors in parallel	$C = C_1 + C_2 + ...$
discharge of a capacitor	$x = x_0 e^{-\frac{t}{RC}}$
Hall voltage	$V_H = \frac{BI}{ntq}$
alternating current/voltage	$x = x_0 \sin \omega t$
radioactive decay	$x = x_0 e^{-\lambda t}$
decay constant	$\lambda = \frac{0.693}{t_{½}}$
intensity reflection coefficient	$\frac{I_R}{I_0} = \frac{(Z_1 - Z_2)^2}{(Z_1 + Z_2)^2}$
Stefan–Boltzmann law	$L = 4\pi\sigma r^2 T^4$
Doppler redshift	$\frac{\Delta\lambda}{\lambda} \approx \frac{\Delta f}{f} \approx \frac{v}{c}$

1 A small bob of mass 0.20 kg is attached to one end of a light inextensible string of length 0.80 m. The other end of the string is attached to a fixed point P. The bob moves with constant speed in a horizontal circle, as shown in Fig. 1.1.

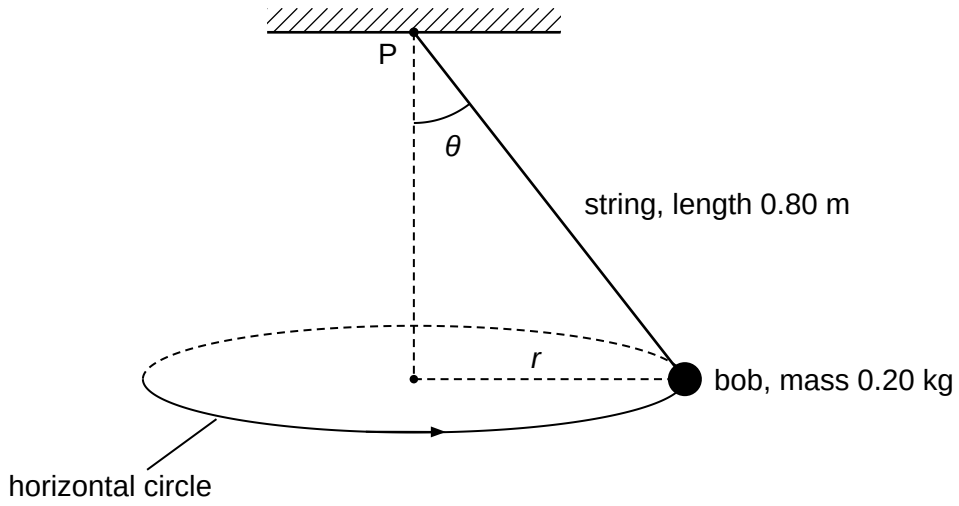


Fig. 1.1

The string makes an angle θ with the vertical. The angular speed of the bob is ω.

(a) Explain why the bob is accelerating even though its speed is constant.

.....

..... [1]

(b) By considering the horizontal component of the tension T in the string, show that

$$T = m\omega^2 L$$

where m is the mass of the bob and L is the length of the string.

[2]

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(c) The string breaks when the tension in it exceeds 12 N. Calculate the maximum angular speed at which the bob can move in this way.

maximum angular speed = ... rad s^{-1} [2]

(d) The angular speed ω of the bob is now varied.

On Fig. 1.2, sketch the variation of cos θ with ω for values of ω from 0 to 8.0 rad s^{-1}.

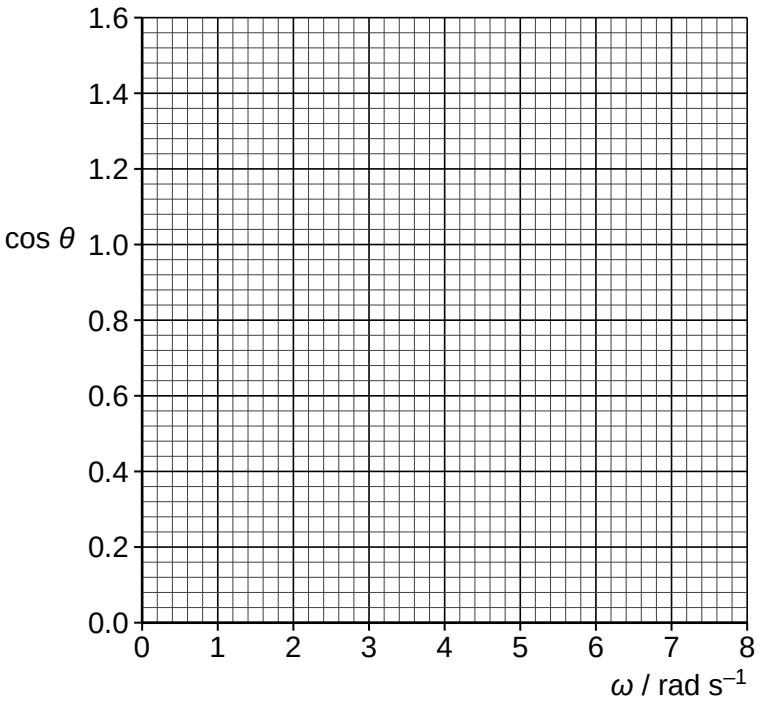


Fig. 1.2

[3]

(e) The bob is replaced by a bob of mass 0.40 kg. The string is unchanged.

State and explain the effect, if any, of this change on the line you have drawn on Fig. 1.2.

.....

.....

..... [2]

[Total: 10]

2 **(a)** Define *gravitational potential* at a point.

.....

.....

..... [2]

(b) The Earth and the Moon may be considered to be isolated uniform spheres. The mass of the Earth is 5.97×10^{24} kg and the mass of the Moon is 7.35×10^{22} kg. The distance between their centres is D, as shown in Fig. 2.1.

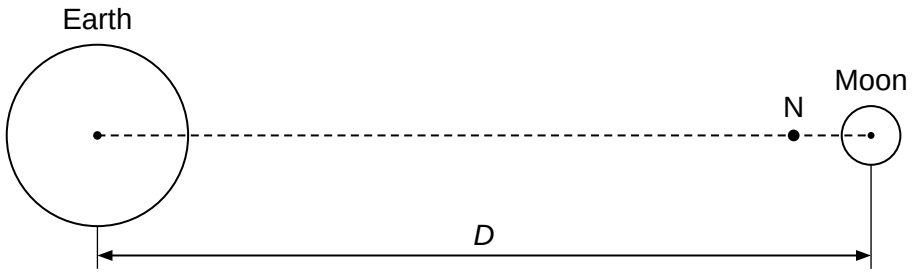


Fig. 2.1 (not to scale)

Point N lies on the line joining the centres of the Earth and the Moon. At point N, the resultant gravitational field strength due to the Earth and the Moon is zero.

(i) Show that the distance of point N from the centre of the Earth is 0.90D.

[2]

(ii) On Fig. 2.2, sketch the variation of gravitational potential φ with distance x from the centre of the Earth, between the surface of the Earth and the surface of the Moon. Point S shows the potential at the surface of the Earth.



Fig. 2.2 (not to scale)

[2]

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(iii) A spacecraft travels from the Earth to the Moon with its engines switched off after launch. Use your answer to (b)(ii) to explain why it needs to be launched with only enough kinetic energy to reach point N.

.....

..... [1]

(c) A satellite of mass 450 kg is in a low circular orbit around the Earth. The upper atmosphere exerts a very small resistive force on the satellite, so that the radius of its orbit slowly decreases. Over several weeks, the radius decreases from 6.72×10^6 m to 6.69×10^6 m. The orbit may be assumed to remain circular.

(i) Determine the change in the kinetic energy of the satellite. State whether the kinetic energy increases or decreases.

change in kinetic energy = ... J [2]

(ii) A student states that, because the resistive force acts in the opposite direction to the velocity of the satellite, the speed of the satellite must decrease. Use your answer to (c)(i) to explain why the student is wrong.

.....

..... [1]

[Total: 10]

3 Two rigid vessels A and B are connected by a narrow tube of negligible volume, as shown in Fig. 3.1.

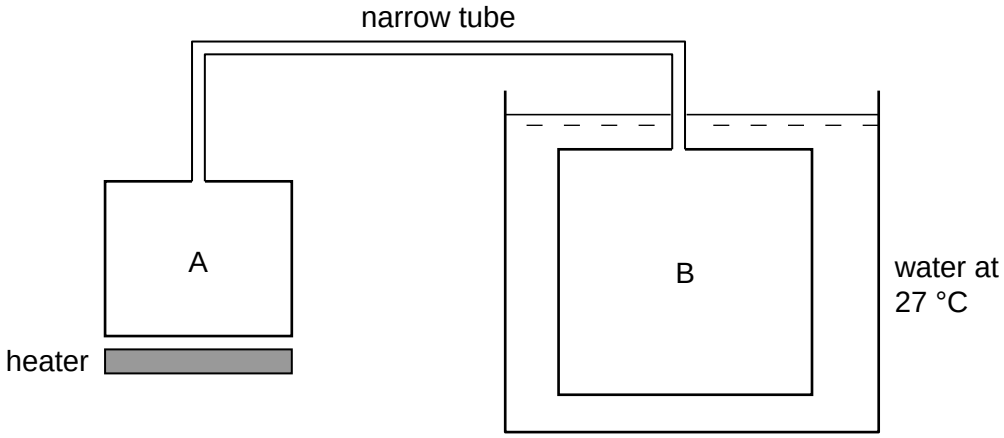


Fig. 3.1

Vessel A has a volume of $1.0 \times 10^{-3}\,m^3$ and vessel B has a volume of $2.0 \times 10^{-3}\,m^3$. The vessels contain an ideal gas. Initially, all of the gas is at a temperature of 27 °C and a pressure of 1.20×10^5 Pa.

Vessel A is then heated so that the gas in it reaches a temperature of 177 °C. Vessel B is kept in a water bath so that the gas in it stays at 27 °C. The gas is left until the pressure no longer changes.

(a) A student states that the temperature of the gas in A has been increased by a factor of 6.6.

Explain why this statement is incorrect.

.....

..... [1]

(b) The student then predicts that the pressure of the gas in A increases by 50% and that the pressure of the gas in B does not change, because the temperature of the gas in B does not change.

Explain why **both** parts of this prediction are wrong.

.....

.....

..... [2]

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(c) **(i)** Show that the final pressure of the gas is 1.35×10^5 Pa.

[2]

(ii) Determine the number of gas molecules that move from A to B.

number = ... [2]

(d) **(i)** The pressure of the gas in B has increased.

State, with a reason, whether the root-mean-square (r.m.s.) speed of the molecules in B has changed.

.....

..... [1]

(ii) Determine the ratio

$$\frac{\text{r.m.s. speed of the molecules in A}}{\text{r.m.s. speed of the molecules in B}}.$$

ratio = ... [2]

[Total: 10]

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4 **(a)** A cylinder fitted with a light frictionless piston contains 0.150 kg of water at 100 °C, as shown in Fig. 4.1.

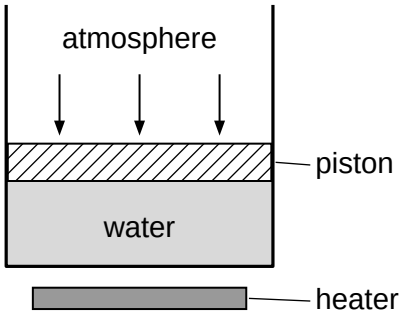


Fig. 4.1

The atmosphere above the piston has a constant pressure of 1.01×10^5 Pa. The water is heated until all of it has turned into steam at 100 °C. The specific latent heat of vaporisation of water is 2.26×10^6 J kg^{-1}. Data for water and steam at 100 °C are given in Table 4.1.

Table 4.1

	water	steam
density / kg m^{-3}	958	0.598

(i) Calculate the work done **on** the water as it turns into steam.

work done on the water = ... J [2]

(ii) Use the first law of thermodynamics to determine the increase in the internal energy of the water as it turns into steam.

increase in internal energy = ... J [2]

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(iii) The temperature of the water does not change as it turns into steam. Explain, in terms of its molecules, why the internal energy of the water increases.

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..... [2]

(b) A fixed mass of an ideal gas is taken around the cycle A → B → C → A shown in Fig. 4.2.

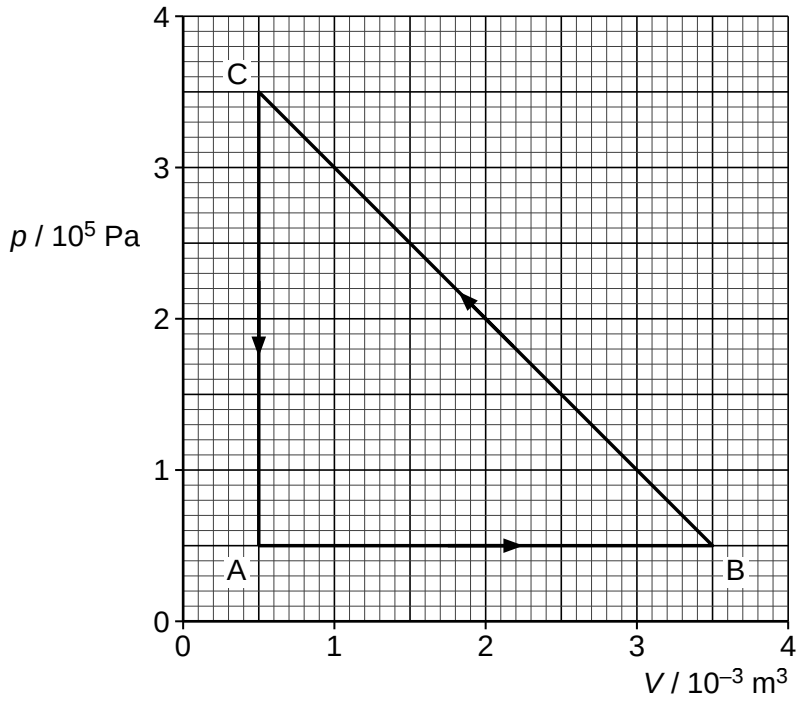


Fig. 4.2

(i) Determine the net work done **on** the gas during one cycle.

net work done on the gas = ... J [2]

(ii) Use the first law of thermodynamics to explain whether there is a net transfer of thermal energy to or from the gas during one cycle. State its magnitude.

.....

.....

..... [2]

[Total: 10]

5 **(a)** State what is meant by *simple harmonic motion*.

.....

.....

..... [2]

(b) A block of mass 0.25 kg hangs from the lower end of a vertical spring. The upper end of the spring is attached to a fixed support, as shown in Fig. 5.1.

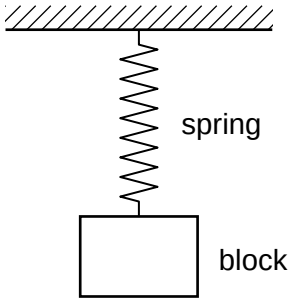


Fig. 5.1

The block is displaced vertically and released. It oscillates with simple harmonic motion. Fig. 5.2 shows part of the variation with time t of the displacement x of the block from its equilibrium position.

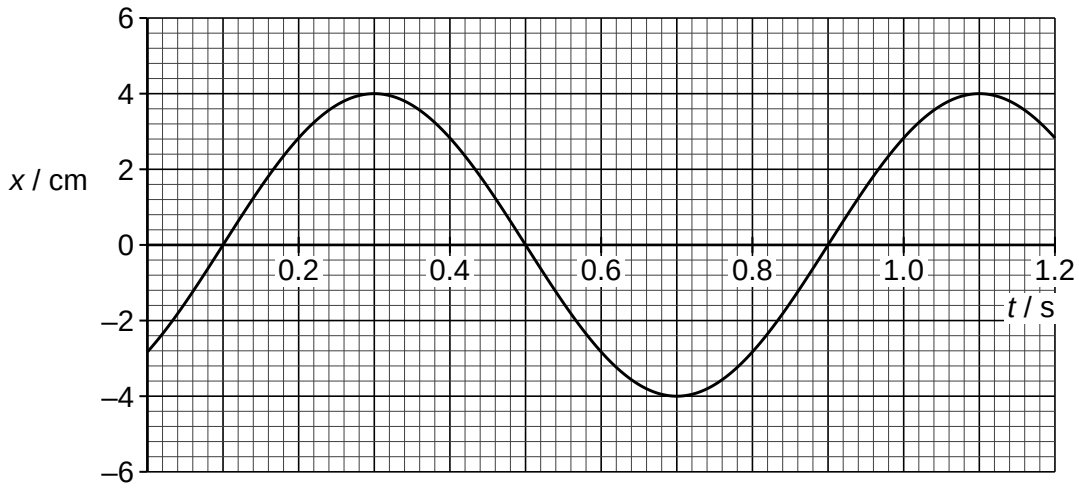


Fig. 5.2

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(i) Use Fig. 5.2 to determine the maximum speed of the block.

maximum speed = ... m s^{-1} [2]

(ii) A student states that, when the displacement of the block is 2.0 cm, half of the total energy of the oscillations is kinetic energy of the block.

By calculating the speed of the block when its displacement is 2.0 cm, determine the fraction of the total energy of the oscillations that is kinetic energy at this displacement.

fraction = ... [3]

(c) The block and spring are taken to the Moon, where the acceleration of free fall is 1.62 m s^{-2}. The block is again made to oscillate vertically. The spring obeys Hooke's law.

A student predicts that the period of the oscillations will be longer on the Moon because the gravitational field strength is smaller.

By considering the resultant force on the block when it is displaced from its equilibrium position, explain why the student is wrong.

.....

.....

.....

..... [3]

[Total: 10]

6 **(a)** Define *capacitance*.

.....

..... [1]

(b) An isolated metal sphere X of radius 0.10 m is supported on an insulating stand, as shown in Fig. 6.1.

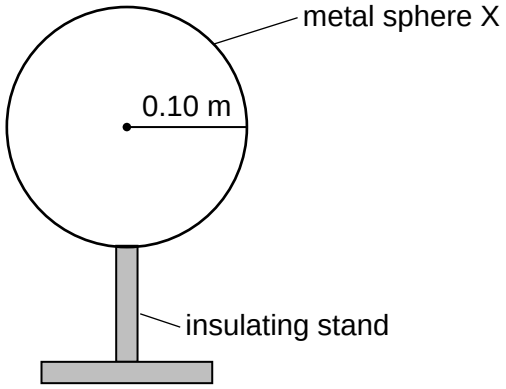


Fig. 6.1

Sphere X is charged so that its potential is 40 kV. The charge on the sphere may be considered to be a point charge at its centre.

Show that the capacitance of sphere X is 1.1×10^{-11} F.

[2]

(c) Sphere X is now connected by a long thin wire to an isolated metal sphere Y of radius 0.050 m. Sphere Y is initially uncharged. The arrangement is shown in Fig. 6.2.

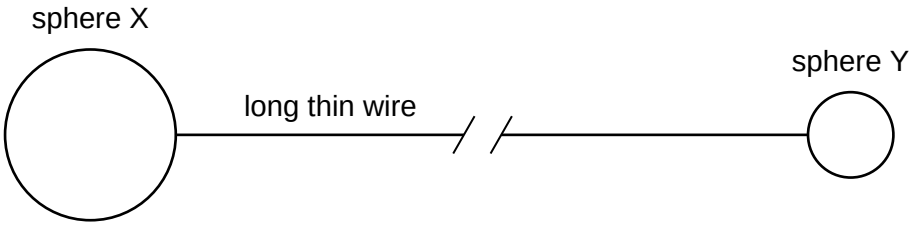


Fig. 6.2 (not to scale)

The spheres are far apart, so that the charge on each sphere has no effect on the potential of the other sphere. Charge flows along the wire until the spheres are at the same potential.

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(i) Determine the final potential of the spheres.

potential = ... kV [2]

(ii) Determine the total energy dissipated as the charge flows along the wire.

energy = ... J [2]

(iii) Suggest what happens to the energy calculated in (c)(ii).

.....

..... [1]

(iv) A student states that, because sphere X has a greater charge than sphere Y, the electric field strength at the surface of X is greater than that at the surface of Y.

Determine the ratio

$$\frac{\text{electric field strength at the surface of Y}}{\text{electric field strength at the surface of X}}.$$

ratio = ... [2]

[Total: 10]

7 **(a)** State what is meant by the *root-mean-square* (r.m.s.) value of an alternating voltage.

.....

.....

..... [2]

(b) The potential difference (p.d.) V across a resistor of resistance 18 Ω varies with time t as shown in Fig. 7.1.

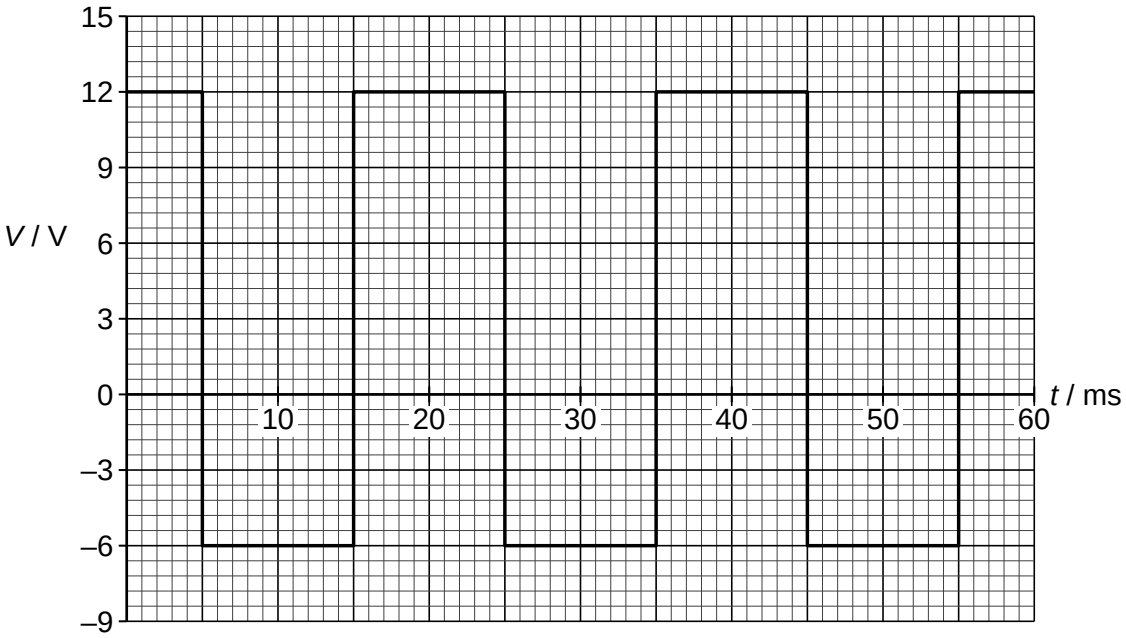


Fig. 7.1

(i) Determine the mean power dissipated in the resistor.

mean power = ... W [2]

(ii) Hence determine the r.m.s. value of V.

r.m.s. value = ... V [1]

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(c) A sinusoidal alternating supply of peak voltage 12 V and frequency 50 Hz is connected to a diode, a capacitor of capacitance 100 μF and a resistor of resistance 1.0 kΩ, as shown in Fig. 7.2.

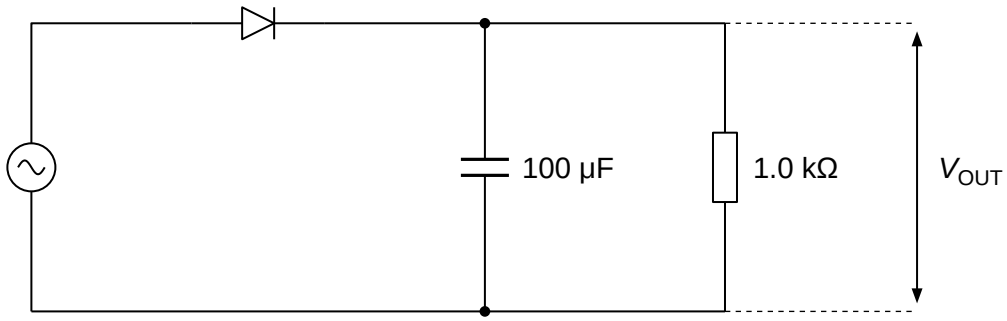


Fig. 7.2

Assume that there is no p.d. across the diode when it is conducting. The output p.d. V_{OUT} varies between a maximum value of 12 V and a minimum value. The difference between the maximum and minimum values is the ripple voltage.

(i) Between successive maximum values of V_{OUT}, the capacitor discharges through the resistor for a time that is approximately equal to one period of the supply.

Determine the ripple voltage.

ripple voltage = ... V [3]

(ii) The circuit is changed so that it produces full-wave rectification. The supply, the capacitor and the resistor are unchanged.

Explain why the ripple voltage is approximately halved.

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..... [2]

[Total: 10]

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8 Fig. 8.1 shows the main parts of a mass spectrometer. The apparatus is in a vacuum.

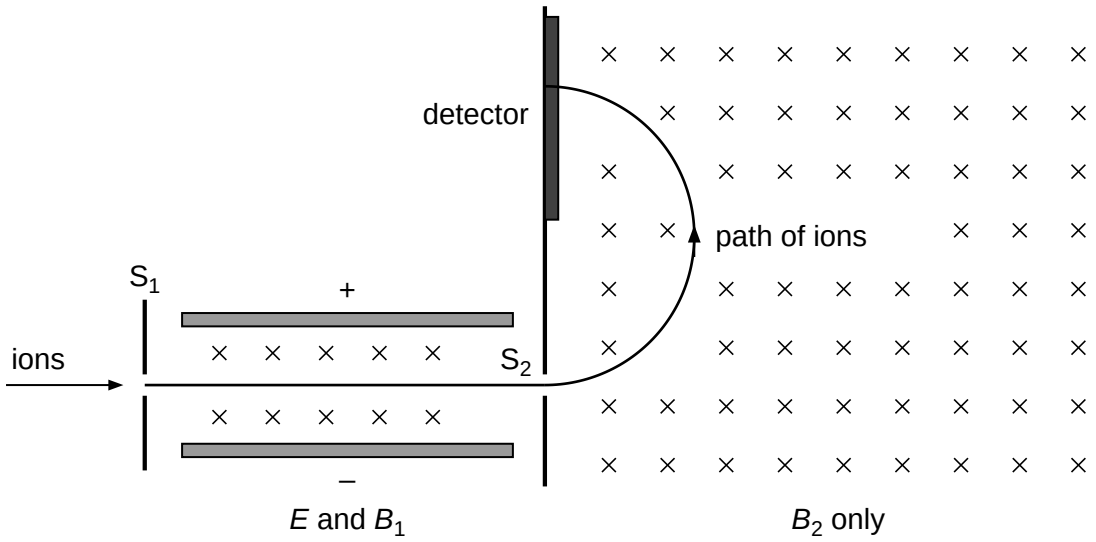


Fig. 8.1

Positive ions pass through slit S_1 into the region between two parallel metal plates, where there is a uniform electric field of strength E and a uniform magnetic field of flux density B_1. Beyond slit S_2 there is a uniform magnetic field of flux density B_2 only. Both magnetic fields are into the page.

(a) Explain why the only ions that pass through slit S_2 are those with speed E / B_1.

.....

.....

..... [2]

(b) The plates are 15 mm apart and the p.d. between them is 1.5 kV. Flux density B_1 is 0.50 T and flux density B_2 is 0.40 T.

A beam of singly charged neon-20 ions ($^{20}Ne^+$) and singly charged neon-22 ions ($^{22}Ne^+$) enters through slit S_1. The mass of each ion, in u, may be taken to be equal to its nucleon number.

(i) Determine the distance between the points at which the two types of ion arrive at the detector.

distance = ... m [3]

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(ii) Some of the neon-22 ions in the beam are doubly charged ($^{22}Ne^{2+}$). Determine the distance from slit S_2 at which these ions arrive at the detector.

distance = ... m [2]

(c) A complete copper ring and an identical copper ring with a narrow slit cut in it are held horizontally above two identical vertical bar magnets, as shown in Fig. 8.2.

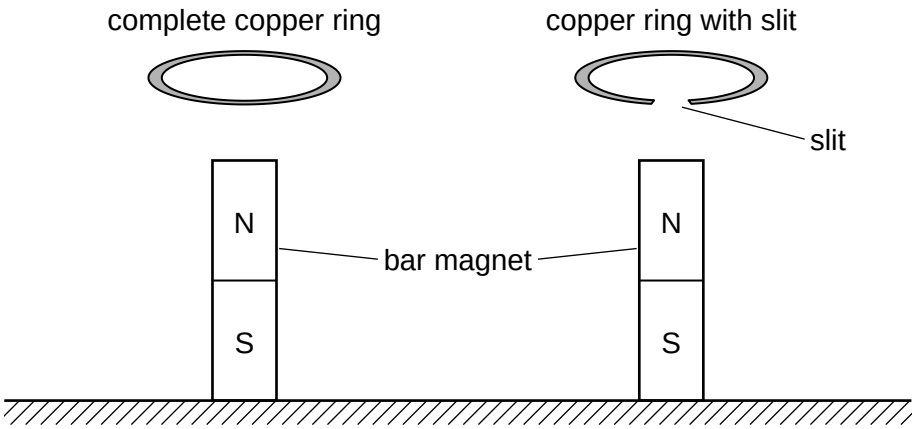


Fig. 8.2

The rings are released at the same time and fall towards the magnets. The ring with the slit falls with acceleration *g*. The complete ring falls with an acceleration that is less than *g*. A student suggests that this is because no electromotive force (e.m.f.) is induced in the ring with the slit.

Explain why the student is wrong, and why the accelerations of the two rings are different.

.....

.....

.....

..... [3]

[Total: 10]

9 **(a)** Electromagnetic radiation of frequency f is incident on the surface of metal X. Fig. 9.1 shows the variation with f of the maximum kinetic energy $E_{K\ max}$ of the electrons emitted from the surface.

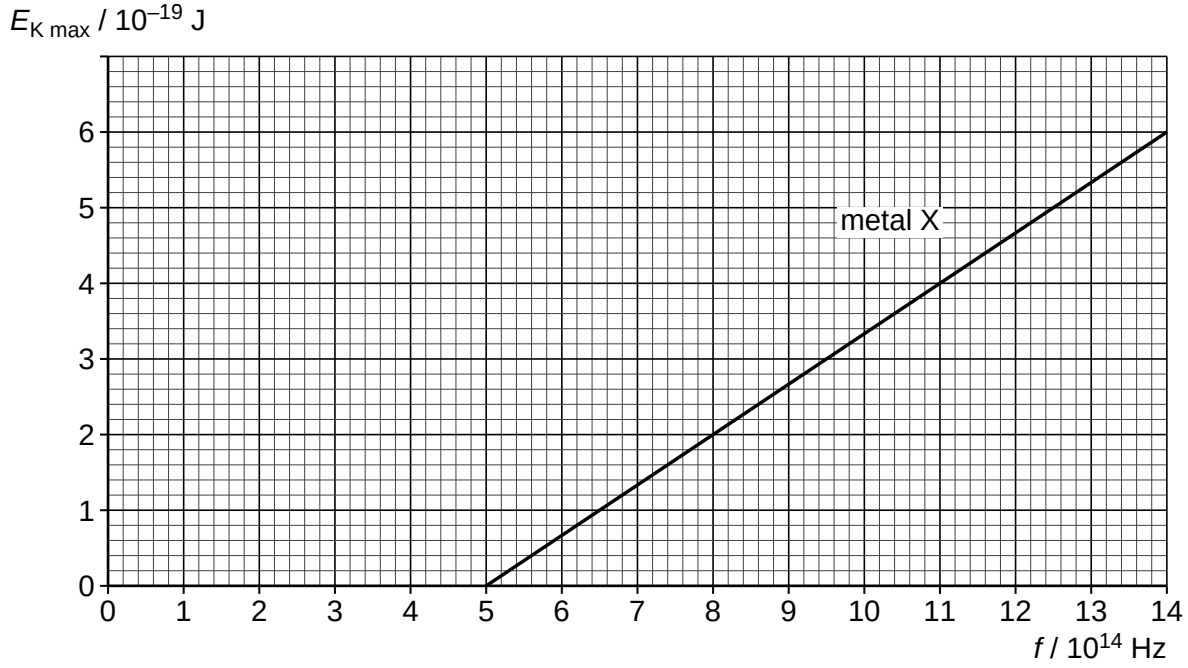


Fig. 9.1

(i) Use Fig. 9.1 to determine a value for the Planck constant.

Planck constant = ... J s [2]

(ii) Determine the work function energy of metal X.

work function energy = ... J [1]

(b) Metal Y has a work function energy of 5.0×10^{-19} J.

On Fig. 9.1, draw a line to show the variation with f of $E_{K\ max}$ for metal Y. [2]

(c) The intensity of the radiation incident on metal X is doubled. The frequency is unchanged.

State and explain the effect, if any, on the line for metal X in Fig. 9.1.

.....

..... [1]

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(d) An electron and a proton are each accelerated from rest through the same potential difference. A student states that, because they gain the same kinetic energy, they have the same de Broglie wavelength.

Determine the ratio

$$\frac{\text{de Broglie wavelength of the electron}}{\text{de Broglie wavelength of the proton}}.$$

ratio = ... [2]

(e) A deuterium nucleus and a tritium nucleus fuse according to the equation

$$^{2}_{1}\text{H} + {}^{3}_{1}\text{H} \rightarrow {}^{4}_{2}\text{He} + {}^{1}_{0}\text{n}$$

Data for the binding energy per nucleon of the nuclei are given in Table 9.1.

Table 9.1

nucleus	binding energy per nucleon / MeV
$^{2}_{1}$H	1.11
$^{3}_{1}$H	2.83
$^{4}_{2}$He	7.07

Determine the energy released, in MeV, in this reaction.

energy = ... MeV [2]

[Total: 10]

10 (a) Ultrasound is used to obtain diagnostic information about structures inside the body. Table 10.1 shows the specific acoustic impedance Z of some materials.

Table 10.1

material	air	gel	soft tissue	bone
Z / kg m^{-2} s^{-1}	4.30×10^2	1.50×10^6	1.63×10^6	7.80×10^6

Use data from Table 10.1 to explain why a gel is applied between the ultrasound transducer and the skin.

.....

.....

..... [2]

(b) A pulse of ultrasound passes through a layer of soft tissue of thickness 6.0 cm and is partly reflected at a boundary between the soft tissue and bone, as shown in Fig. 10.1.

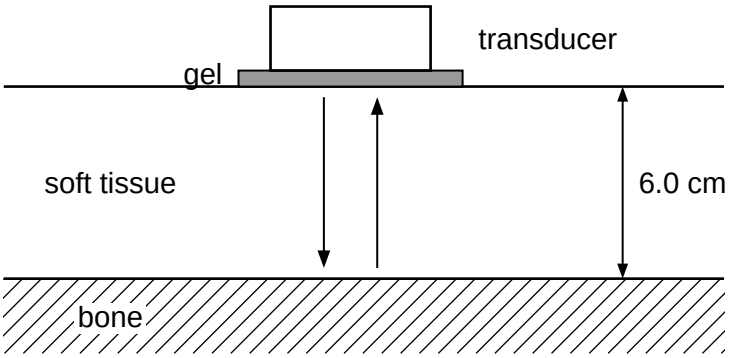


Fig. 10.1

The reflected pulse returns through the same soft tissue to the transducer. The linear attenuation coefficient of soft tissue for this ultrasound is 0.23 cm^{-1}. Assume that there is no reflection at any other boundary.

Calculate the intensity of the reflected pulse arriving at the transducer as a fraction of the intensity of the pulse entering the soft tissue.

fraction = ... [3]

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(c) Fig. 10.2 shows the variation with wavelength λ of the intensity of the radiation emitted by two stars P and Q. Each curve has been scaled so that its maximum value is 1.0.

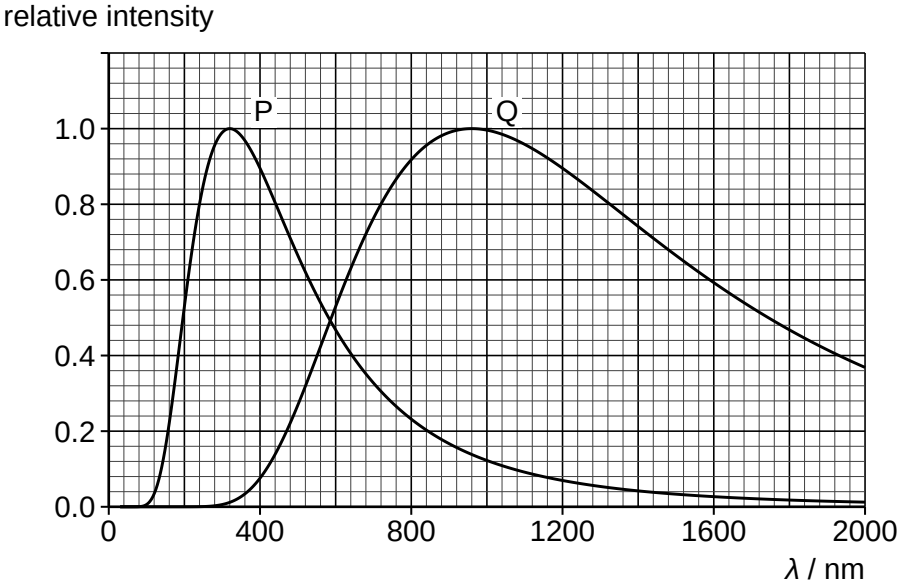


Fig. 10.2

The radius of star Q is 45 times the radius of star P. A student states that star P must have the greater luminosity because it has the higher surface temperature. Use Fig. 10.2 to determine the ratio of the luminosity of star Q to the luminosity of star P.

ratio = ... [3]

(d) A spectral line of hydrogen has a wavelength of 656.3 nm when measured in a laboratory. In the light from a distant galaxy, the same line has a wavelength of 678.9 nm. The galaxy is a distance of 4.6×10^{24} m from the Earth.

Assuming that the galaxy has moved away from the Earth at a constant speed since the Big Bang, estimate the age of the universe, in years.

age = ... years [2]

[Total: 10]



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